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Topik 9 - Recursion

1. Problem

Create a recursive function and program for the following equation!

- f(5) = 5*(4/2)*3*(2/2)*1
- f(7) = 7*(6/2)*5*(4/2)*3*(2/2)*1

You must create them using a recursive approach, and provide an explanation on top of the program using comments.

2. Solution

- Looking at f(7) we will find that there is a f(5) problem inside f(7), so because every problem has a smaller sub problem than function f is a recursive function.
- Looking at f(5) we can rewrite it into 5 * f(4) and f(4) we can rewrite as (4/2) * f(3). Looking at this pattern we can create two recurrent, that is:
  - When n > 1 and n is even then return (n/2) * f(n - 1).
  - When n > 1 and n is odd then return n * f(n - 1).
- The last step is the base case, looking f(5) and f(7) both of that function end at 1, so it is safe to assume that the base case is when n is equal to 1 then return 1.
- So the recursive function is:
  - Return 1 when n is equal to 1
  - Return (n/2) * f(n - 1) when n > 1 and n is even
  - Return n * f(n - 1) when n > 1 and n is odd

3. Code

```cpp
// nama: Jihad Malika Rahman
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#include <iostream>
using namespace std;

int f(int n){
    if(n <= 1){
        return 1; // (base case)
    }else if(n % 2 == 0){
        return f(n - 1) * (n / 2); // (recurrent)
    }else{
        return f(n - 1) * n; // (recurrent)
    }
}

int main(){
    int n;
    cin >> n;
    cout << f(n);
}
```